

Assignment 7 – API Gateway and Cognito

Updated demo recording: [SNS SQS Serverless demo AI.mp4](#)

Upload screenshots of:

- API Gateway resources you defined. There should be all CRUD endpoints.
- Cognito users in the user pool
- Authorizer
- Token
- Sending requests with a Postman with an Authorization header.

If you get a CORS error, add the following to the lambda response:

```
headers: {  
  'Access-Control-Allow-Origin': '*',  
  'Access-Control-Allow-Headers': 'Content-Type,Authorization,X-Amz-Date,X-Api-Key',  
  'Access-Control-Allow-Methods': 'DELETE,GET,HEAD,OPTIONS,PATCH,POST,PUT'  
},
```

PART I – Lambda and DynamoDB [is already done in the previous assignments]

In the previous lab, we prepared a CRUD app with Lambda and DynamoDB. In this lab, we will make that a real API. So, it is publicly accessible. We will also secure it with JWT tokens from Cognito.

PART II - API gateway and Cognito

Task 1. API Gateway

- Create a CRUD API for the sample course app in API Gateway.
 - Create a REST API. The process is straightforward. Only give it a name. Create a resource called “courses”.
 - Create a “method” in it.
 - Select a method such as POST, GET, PATCH.
 - You must toggle **Lambda proxy integration**
 - Select the Lambda function that has the implementation.
- Create the following 5 endpoints:

/courses POST – It will save a course record and return 201 status.

/courses GET – It will return all items in the table if there are no query parameters. If there are query parameters, return records that match the query string criteria. Under the hood, it will use the “scan” function with/without “filter-expression”.

/courses/{courseCode}?teacherName={value} GET – It will return a single item that matches the “courseCode” in the path param and the “teacherName” in the query string. Its implementation is the “getItem” method of DynamoDB API.

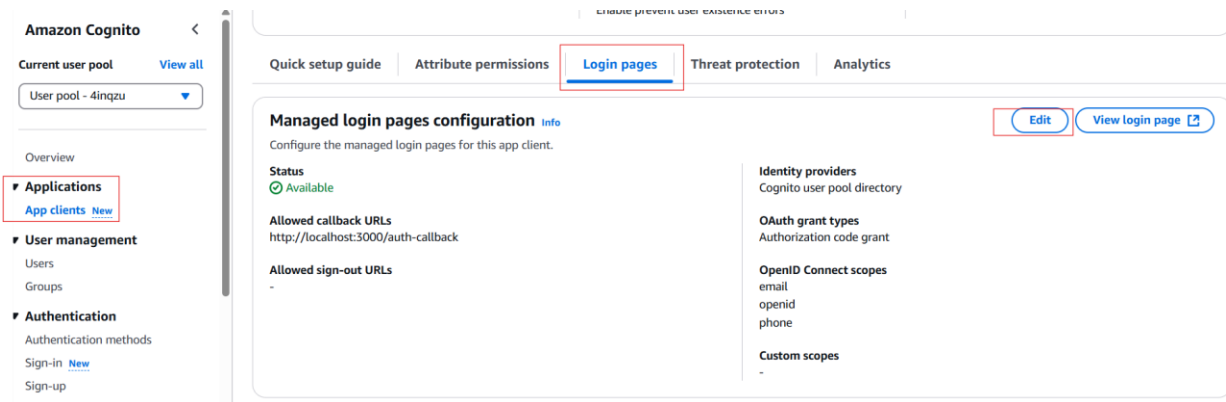
/courses PATCH – Updates an existing item that calls the “updateItem” implementation in your lambda.

/courses DELETE – Deletes an existing item that calls the “deleteItem” implementation in your lambda.

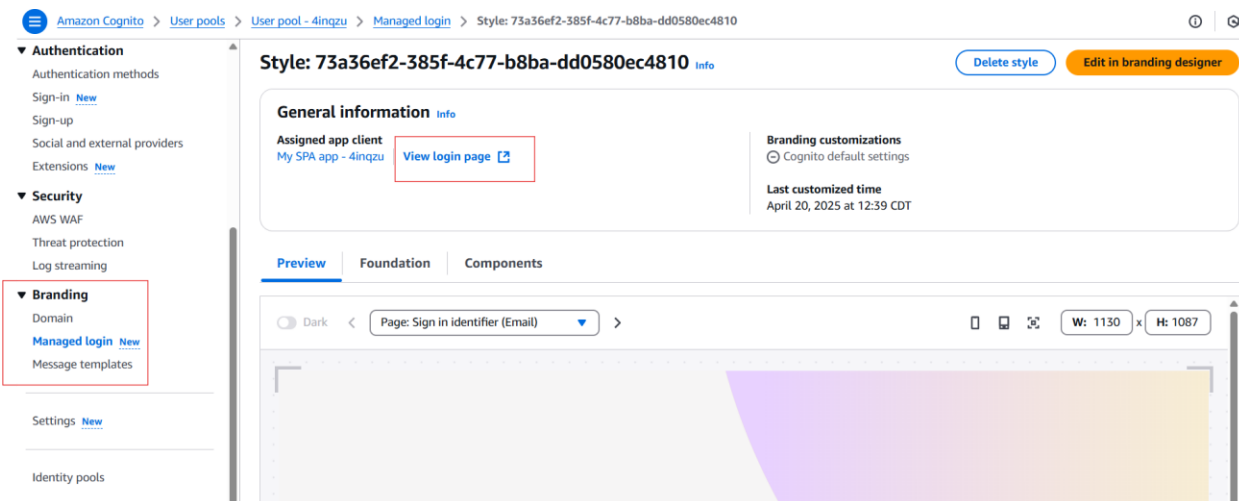
Task 2. Cognito

Create a **Cognito User pool** for the app. The main configs are, enabling the hosted UI to get tokens and selecting IMPLICIT GRANT. Step by step:

- Create a user pool.
 - Select Single-page application (SPA)
 - Add a return URL, <http://localhost:3000>
- Go to the user pool and select the App Client. Click on the Login pages table. Hit edit.



- Remove Authorization code grant in OAuth 2.0 grant types. Instead, select **Implicit grant**.
- Go to Managed Logins. Click on the Style ID. And click on View login page.



- Create a user in the managed login page. Login with the user and you will see the tokens in the hash parameter. Grab only ID token.

Task 3. Secure the API using tokens from the Cognito User pool.

- Go to Authorizers in the API
- Give it a name. Select Cognito as Authorizer type. Select to user pool. The token source is “Authorization”.

- Go to the resources. Click on the method and click on the “Method request”. Click “Edit” in the Method request settings. In the Authorization select the authorizer. **Deploy** the API.

Task4. Test the API

1. Go to the Cognito pool and click on “App integration”. Scroll all the way down and click on the app client. Click on the “View Hosted UI” in the hosted UI section. Sign up and enter the code in your email. It will redirect you to the callback URL you entered and in the URL you will see **id_token** as a hash parameter.
2. Copy the **ID Token**. Provide it in the header as **Authorization**.

The screenshot shows a REST client interface with the following details:

- URL:** `https://koh7g6lhle.execute-api.us-east-1.amazonaws.com/v1`
- Method:** `POST`
- Headers (9):**

Key	Value	Description
☒ Authorization	eyJraWQiOiJEanZ2czFtc0prSXNFWndhbnJrSXN1XC85d...	
Key	Value	Description

Common errors

- **403 Forbidden** - You are hitting an endpoint that is not defined in the resource. For example, you only have a GET endpoint defined in the API Gateway but trying to access the POST endpoint.
- **502 Bad Gateway** – Your lambda should return a proper JSON that API Gateway understands.